

follows from equation (B.3) on page 1156. For example, if each position on a car's license plate is a letter or a digit, then the number of possibilities for each position is $26 + 10 = 36$, since there are 26 choices if it is a letter and 10 choices if it is a digit.

The **rule of product** says that the number of ways to choose an ordered pair is the number of ways to choose the first element times the number of ways to choose the second element. That is, if A and B are two finite sets, then $|A \times B| = |A| \cdot |B|$, which is simply equation (B.4) on page 1157. For example, if an ice-cream parlor offers 28 flavors of ice cream and four toppings, the number of possible sundaes with one scoop of ice cream and one topping is $28 \cdot 4 = 112$.

Strings

A **string** over a finite set S is a sequence of elements of S . For example, there are eight binary strings of length 3:

000, 001, 010, 011, 100, 101, 110, 111 .

(Here we use the shorthand of omitting the angle brackets when denoting a sequence.) We sometimes call a string of length k a **k -string**. A **substring** s' of a string s is an ordered sequence of consecutive elements of s . A **k -substring** of a string is a substring of length k . For example, 010 is a 3-substring of 01101001 (the 3-substring that begins in position 4), but 111 is not a substring of 01101001.

We can view a k -string over a set S as an element of the Cartesian product S^k of k -tuples, which means that there are $|S|^k$ strings of length k . For example, the number of binary k -strings is 2^k . Intuitively, to construct a k -string over an n -set, there are n ways to pick the first element; for each of these choices, there are n ways to pick the second element; and so forth k times. This construction leads to the k -fold product $\underbrace{n \cdot n \cdots n}_{n \text{ times}} = n^k$ as the number of k -strings.

Permutations

A **permutation** of a finite set S is an ordered sequence of all the elements of S , with each element appearing exactly once. For example, if $S = \{a, b, c\}$, then S has 6 permutations:

$abc, acb, bac, bca, cab, cba$.

(Again, we use the shorthand of omitting the angle brackets when denoting a sequence.) There are $n!$ permutations of a set of n elements, since there are n ways to choose the first element of the sequence, $n - 1$ ways for the second element, $n - 2$ ways for the third, and so on.

A ***k*-permutation** of S is an ordered sequence of k elements of S , with no element appearing more than once in the sequence. (Thus, an ordinary permutation is an n -permutation of an n -set.) Here are the 2-permutations of the set $\{a, b, c, d\}$:

$ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db, dc$.

The number of k -permutations of an n -set is

$$n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!} , \quad (\text{C.1})$$

since there are n ways to choose the first element, $n-1$ ways to choose the second element, and so on, until k elements are chosen, with the last element chosen from the remaining $n-k+1$ elements. For the above example, with $n=4$ and $k=2$, the formula (C.1) evaluates to $4!/2! = 12$, matching the number of 2-permutations listed.

Combinations

A ***k*-combination** of an n -set S is simply a k -subset of S . For example, the 4-set $\{a, b, c, d\}$ has six 2-combinations:

ab, ac, ad, bc, bd, cd .

(Here we use the shorthand of omitting the braces around each subset.) To construct a k -combination of an n -set, choose k distinct (different) elements from the n -set. The order of selecting the elements does not matter.

We can express the number of k -combinations of an n -set in terms of the number of k -permutations of an n -set. Every k -combination has exactly $k!$ permutations of its elements, each of which is a distinct k -permutation of the n -set. Thus the number of k -combinations of an n -set is the number of k -permutations divided by $k!$. From equation (C.1), this quantity is

$$\frac{n!}{k!(n-k)!} . \quad (\text{C.2})$$

For $k=0$, this formula tells us that the number of ways to choose 0 elements from an n -set is 1 (not 0), since $0! = 1$.

Binomial coefficients

The notation $\binom{n}{k}$ (read “ n choose k ”) denotes the number of k -combinations of an n -set. Equation (C.2) gives

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} .$$

This formula is symmetric in k and $n - k$:

$$\binom{n}{k} = \binom{n}{n-k}. \quad (\text{C.3})$$

These numbers are also known as *binomial coefficients*, due to their appearance in the *binomial theorem*:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}, \quad (\text{C.4})$$

where $n \in \mathbb{N}$ and $x, y \in \mathbb{R}$. The right-hand side of equation (C.4) is called the *binomial expansion* of the left-hand side. A special case of the binomial theorem occurs when $x = y = 1$:

$$2^n = \sum_{k=0}^n \binom{n}{k}.$$

This formula corresponds to counting the 2^n binary n -strings by the number of 1s they contain: $\binom{n}{k}$ binary n -strings contain exactly k 1s, since there are $\binom{n}{k}$ ways to choose k out of the n positions in which to place the 1s.

Many identities involve binomial coefficients. The exercises at the end of this section give you the opportunity to prove a few.

Binomial bounds

You sometimes need to bound the size of a binomial coefficient. For $1 \leq k \leq n$, we have the lower bound

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1) \cdots (n-k+1)}{k(k-1) \cdots 1} \\ &= \left(\frac{n}{k}\right) \left(\frac{n-1}{k-1}\right) \cdots \left(\frac{n-k+1}{1}\right) \\ &\geq \left(\frac{n}{k}\right)^k. \end{aligned} \quad (\text{C.5})$$

Taking advantage of the inequality $k! \geq (k/e)^k$ derived from Stirling's approximation (3.25) on page 67, we obtain the upper bounds

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1) \cdots (n-k+1)}{k(k-1) \cdots 1} \\ &\leq \frac{n^k}{k!} \\ &\leq \left(\frac{en}{k}\right)^k. \end{aligned} \quad (\text{C.6})$$

For all integers k such that $0 \leq k \leq n$, you can use induction (see Exercise C.1-12) to prove the bound

$$\binom{n}{k} \leq \frac{n^n}{k^k (n-k)^{n-k}}, \quad (\text{C.7})$$

where for convenience we assume that $0^0 = 1$. For $k = \lambda n$, where $0 \leq \lambda \leq 1$, we can rewrite this bound as

$$\begin{aligned} \binom{n}{\lambda n} &\leq \frac{n^n}{(\lambda n)^{\lambda n} ((1-\lambda)n)^{(1-\lambda)n}} \\ &= \left(\left(\frac{1}{\lambda} \right)^\lambda \left(\frac{1}{1-\lambda} \right)^{1-\lambda} \right)^n \\ &= 2^{n H(\lambda)}, \end{aligned}$$

where

$$H(\lambda) = -\lambda \lg \lambda - (1-\lambda) \lg(1-\lambda) \quad (\text{C.8})$$

is the **(binary) entropy function** and where, for convenience, we assume that $0 \lg 0 = 0$, so that $H(0) = H(1) = 0$.

Exercises

C.1-1

How many k -substrings does an n -string have? (Consider identical k -substrings at different positions to be different.) How many substrings does an n -string have in total?

C.1-2

An n -input, m -output **boolean function** is a function from $\{0, 1\}^n$ to $\{0, 1\}^m$. How many n -input, 1-output boolean functions are there? How many n -input, m -output boolean functions are there?

C.1-3

In how many ways can n professors sit around a circular conference table? Consider two seatings to be the same if one can be rotated to form the other.

C.1-4

In how many ways is it possible to choose three distinct numbers from the set $\{1, 2, \dots, 99\}$ so that their sum is even?

C.1-5

Prove the identity

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1} \quad (\text{C.9})$$

for $0 < k \leq n$.

C.1-6

Prove the identity

$$\binom{n}{k} = \frac{n}{n-k} \binom{n-1}{k}$$

for $0 \leq k < n$.

C.1-7

To choose k objects from n , you can make one of the objects distinguished and consider whether the distinguished object is chosen. Use this approach to prove that

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

C.1-8

Using the result of Exercise C.1-7, make a table for $n = 0, 1, \dots, 6$ and $0 \leq k \leq n$ of the binomial coefficients $\binom{n}{k}$ with $\binom{0}{0}$ at the top, $\binom{1}{0}$ and $\binom{1}{1}$ on the next line, then $\binom{2}{0}$, $\binom{2}{1}$, and $\binom{2}{2}$, and so forth. Such a table of binomial coefficients is called *Pascal's triangle*.

C.1-9

Prove that

$$\sum_{i=1}^n i = \binom{n+1}{2}.$$

C.1-10

Show that for any integers $n \geq 0$ and $0 \leq k \leq n$, the expression $\binom{n}{k}$ achieves its maximum value when $k = \lfloor n/2 \rfloor$ or $k = \lceil n/2 \rceil$.

★ **C.1-11**

Argue that for any integers $n \geq 0$, $j \geq 0$, $k \geq 0$, and $j + k \leq n$,

$$\binom{n}{j+k} \leq \binom{n}{j} \binom{n-j}{k}. \quad (\text{C.10})$$

Provide both an algebraic proof and an argument based on a method for choosing $j + k$ items out of n . Give an example in which equality does not hold.

★ **C.1-12**

Use induction on all integers k such that $0 \leq k \leq n/2$ to prove inequality (C.7), and use equation (C.3) to extend it to all integers k such that $0 \leq k \leq n$.

★ **C.1-13**

Use Stirling's approximation to prove that

$$\binom{2n}{n} = \frac{2^{2n}}{\sqrt{\pi n}} (1 + O(1/n)). \quad (\text{C.11})$$

★ **C.1-14**

By differentiating the entropy function $H(\lambda)$, show that it achieves its maximum value at $\lambda = 1/2$. What is $H(1/2)$?

★ **C.1-15**

Show that for any integer $n \geq 0$,

$$\sum_{k=0}^n \binom{n}{k} k = n 2^{n-1}. \quad (\text{C.12})$$

★ **C.1-16**

Inequality (C.5) provides a lower bound on the binomial coefficient $\binom{n}{k}$. For small values of k , a stronger bound holds. Prove that

$$\binom{n}{k} \geq \frac{n^k}{4k!} \quad (\text{C.13})$$

for $k \leq \sqrt{n}$.

C.2 Probability

Probability is an essential tool for the design and analysis of probabilistic and randomized algorithms. This section reviews basic probability theory.

We define probability in terms of a *sample space* S , which is a set whose elements are called *outcomes* or *elementary events*. Think of each outcome as a possible result of an experiment. For the experiment of flipping two distinguishable coins, with each individual flip resulting in a head (H) or a tail (T), you can view the sample space S as consisting of the set of all possible 2-strings over $\{H, T\}$:

$$S = \{HH, HT, TH, TT\} .$$

An *event* is a subset¹ of the sample space S . For example, in the experiment of flipping two coins, the event of obtaining one head and one tail is $\{HT, TH\}$. The event S is called the *certain event*, and the event $\emptyset$ is called the *null event*. We say that two events A and B are *mutually exclusive* if $A \cap B = \emptyset$. An outcome s also defines the event $\{s\}$, which we sometimes write as just s . By definition, all outcomes are mutually exclusive.

Axioms of probability

A *probability distribution* $\Pr\{\}$ on a sample space S is a mapping from events of S to real numbers satisfying the following *probability axioms*:

1. $\Pr\{A\} \geq 0$ for any event A .
2. $\Pr\{S\} = 1$.
3. $\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\}$ for any two mutually exclusive events A and B .
More generally, for any sequence of events $A_1, A_2, \dots$ (finite or countably infinite) that are pairwise mutually exclusive,

$$\Pr\left\{\bigcup_i A_i\right\} = \sum_i \Pr\{A_i\} .$$

We call $\Pr\{A\}$ the *probability* of the event A . Axiom 2 is simply a normalization requirement: there is really nothing fundamental about choosing 1 as the probability of the certain event, except that it is natural and convenient.

Several results follow immediately from these axioms and basic set theory (see Section B.1). The null event $\emptyset$ has probability $\Pr\{\emptyset\} = 0$. If $A \subseteq B$, then

¹ For a general probability distribution, there may be some subsets of the sample space S that are not considered to be events. This situation usually arises when the sample space is uncountably infinite. The main requirement for what subsets are events is that the set of events of a sample space must be closed under the operations of taking the complement of an event, forming the union of a finite or countable number of events, and taking the intersection of a finite or countable number of events. Most of the probability distributions we see in this book are over finite or countable sample spaces, and we generally consider all subsets of a sample space to be events. A notable exception is the continuous uniform probability distribution, which we'll see shortly.

$\Pr\{A\} \leq \Pr\{B\}$. Using $\overline{A}$ to denote the event $S - A$ (the *complement* of A), we have $\Pr\{\overline{A}\} = 1 - \Pr\{A\}$. For any two events A and B ,

$$\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\} - \Pr\{A \cap B\} \quad (\text{C.14})$$

$$\leq \Pr\{A\} + \Pr\{B\} . \quad (\text{C.15})$$

In our coin-flipping example, suppose that each of the four outcomes has probability $1/4$. Then the probability of getting at least one head is

$$\begin{aligned} \Pr\{HH, HT, TH\} &= \Pr\{HH\} + \Pr\{HT\} + \Pr\{TH\} \\ &= 3/4 . \end{aligned}$$

Another way to obtain the same result is to observe that since the probability of getting strictly less than one head is $\Pr\{TT\} = 1/4$, the probability of getting at least one head is $1 - 1/4 = 3/4$.

Discrete probability distributions

A probability distribution is *discrete* if it is defined over a finite or countably infinite sample space. Let S be the sample space. Then for any event A ,

$$\Pr\{A\} = \sum_{s \in A} \Pr\{s\} ,$$

since outcomes, specifically those in A , are mutually exclusive. If S is finite and every outcome $s \in S$ has probability $\Pr\{s\} = 1/|S|$, then we have the *uniform probability distribution* on S . In such a case the experiment is often described as “picking an element of S at random.”

As an example, consider the process of flipping a *fair coin*, one for which the probability of obtaining a head is the same as the probability of obtaining a tail, that is, $1/2$. Flipping the coin n times gives the uniform probability distribution defined on the sample space $S = \{H, T\}^n$, a set of size 2^n . We can represent each outcome in S as a string of length n over $\{H, T\}$, with each string occurring with probability $1/2^n$. The event $A = \{\text{exactly } k \text{ heads and exactly } n - k \text{ tails occur}\}$ is a subset of S of size $|A| = \binom{n}{k}$, since $\binom{n}{k}$ strings of length n over $\{H, T\}$ contain exactly k H's. The probability of event A is thus $\Pr\{A\} = \binom{n}{k}/2^n$.

Continuous uniform probability distribution

The continuous uniform probability distribution is an example of a probability distribution in which not all subsets of the sample space are considered to be events. The continuous uniform probability distribution is defined over a closed interval $[a, b]$ of the reals, where $a < b$. The intuition is that each point in the interval $[a, b]$ should be “equally likely.” Because there are an uncountable number

of points, however, if all points had the same finite, positive probability, axioms 2 and 3 would not be simultaneously satisfied. For this reason, we'd like to associate a probability only with *some* of the subsets of S in such a way that the axioms are satisfied for these events.

For any closed interval $[c, d]$, where $a \leq c \leq d \leq b$, the **continuous uniform probability distribution** defines the probability of the event $[c, d]$ to be

$$\Pr\{[c, d]\} = \frac{d - c}{b - a}.$$

Letting $c = d$ gives that the probability of a single point is 0. Removing the endpoints $[c, c]$ and $[d, d]$ of an interval $[c, d]$ results in the open interval (c, d) . Since $[c, d] = [c, c] \cup (c, d) \cup [d, d]$, axiom 3 gives $\Pr\{[c, d]\} = \Pr\{(c, d)\}$. Generally, the set of events for the continuous uniform probability distribution contains any subset of the sample space $[a, b]$ that can be obtained by a finite or countable union of open and closed intervals, as well as certain more complicated sets.

Conditional probability and independence

Sometimes you have some prior partial knowledge about the outcome of an experiment. For example, suppose that a friend has flipped two fair coins and has told you that at least one of the coins showed a head. What is the probability that both coins are heads? The information given eliminates the possibility of two tails. The three remaining outcomes are equally likely, and so you infer that each occurs with probability $1/3$. Since only one of these outcomes shows two heads, the answer is $1/3$.

Conditional probability formalizes the notion of having prior partial knowledge of the outcome of an experiment. The **conditional probability** of an event A given that another event B occurs is defined to be

$$\Pr\{A \mid B\} = \frac{\Pr\{A \cap B\}}{\Pr\{B\}} \quad (\text{C.16})$$

whenever $\Pr\{B\} \neq 0$. (Read " $\Pr\{A \mid B\}$ " as "the probability of A given B .") The idea behind equation (C.16) is that since we are given that event B occurs, the event that A also occurs is $A \cap B$. That is, $A \cap B$ is the set of outcomes in which both A and B occur. Because the outcome is one of the elementary events in B , we normalize the probabilities of all the elementary events in B by dividing them by $\Pr\{B\}$, so that they sum to 1. The conditional probability of A given B is, therefore, the ratio of the probability of event $A \cap B$ to the probability of event B . In the example above, A is the event that both coins are heads, and B is the event that at least one coin is a head. Thus, $\Pr\{A \mid B\} = (1/4)/(3/4) = 1/3$.

Two events are *independent* if

$$\Pr\{A \cap B\} = \Pr\{A\} \Pr\{B\} , \quad (\text{C.17})$$

which is equivalent, if $\Pr\{B\} \neq 0$, to the condition

$$\Pr\{A \mid B\} = \Pr\{A\} .$$

For example, suppose that you flip two fair coins and that the outcomes are independent. Then the probability of two heads is $(1/2)(1/2) = 1/4$. Now suppose that one event is that the first coin comes up heads and the other event is that the coins come up differently. Each of these events occurs with probability $1/2$, and the probability that both events occur is $1/4$. Thus, according to the definition of independence, the events are independent—even though you might think that both events depend on the first coin. Finally, suppose that the coins are welded together so that they both fall heads or both fall tails and that the two possibilities are equally likely. Then the probability that each coin comes up heads is $1/2$, but the probability that they both come up heads is $1/2 \neq (1/2)(1/2)$. Consequently, the event that one comes up heads and the event that the other comes up heads are not independent.

A collection $A_1, A_2, \dots, A_n$ of events is said to be *pairwise independent* if

$$\Pr\{A_i \cap A_j\} = \Pr\{A_i\} \Pr\{A_j\}$$

for all $1 \leq i < j \leq n$. We say that the events of the collection are *(mutually) independent* if every k -subset $A_{i_1}, A_{i_2}, \dots, A_{i_k}$ of the collection, where $2 \leq k \leq n$ and $1 \leq i_1 < i_2 < \dots < i_k \leq n$, satisfies

$$\Pr\{A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}\} = \Pr\{A_{i_1}\} \Pr\{A_{i_2}\} \dots \Pr\{A_{i_k}\} .$$

For example, suppose that you flip two fair coins. Let A_1 be the event that the first coin is heads, let A_2 be the event that the second coin is heads, and let A_3 be the event that the two coins are different. Then,

$$\Pr\{A_1\} = 1/2 ,$$

$$\Pr\{A_2\} = 1/2 ,$$

$$\Pr\{A_3\} = 1/2 ,$$

$$\Pr\{A_1 \cap A_2\} = 1/4 ,$$

$$\Pr\{A_1 \cap A_3\} = 1/4 ,$$

$$\Pr\{A_2 \cap A_3\} = 1/4 ,$$

$$\Pr\{A_1 \cap A_2 \cap A_3\} = 0 .$$

Since for $1 \leq i < j \leq 3$, we have $\Pr\{A_i \cap A_j\} = \Pr\{A_i\} \Pr\{A_j\} = 1/4$, the events A_1, A_2 , and A_3 are pairwise independent. The events are not mutually independent, however, because $\Pr\{A_1 \cap A_2 \cap A_3\} = 0$ and $\Pr\{A_1\} \Pr\{A_2\} \Pr\{A_3\} = 1/8 \neq 0$.